SYMBOLIC POWERS ARE PRESERVED BY A FLAT EXTENSION ALONG WHICH THE PRIME STAYS PRIME

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Allowed standard prerequisites (editorial).

- Symbolic power as the kernel of a map to a localised quotient. Used at the start of the counted proof (line 15803). Represents the n-th symbolic power as a kernel, which flatness then transports.
- Filtrations with subquotients given by tensor products. Used near the end of the counted proof (lines 15820-15827). Shows that the elements outside the prime act invertibly, which gives the injectivity needed.

Lemma 0.1. *Let $R \to S$ be flat ring map. Let $\mathfrak{p} \subset R$ be a prime such that $\mathfrak{q} = \mathfrak{p}S$ is a prime of S. Then $\mathfrak{p}^{(n)}S = \mathfrak{q}^{(n)}$.*

Proof. Since $\mathfrak{p}^{(n)} = \mathrm{Ker}(R \to R_\mathfrak{p}/\mathfrak{p}^n R_\mathfrak{p})$ we see using flatness that $\mathfrak{p}^{(n)}S$ is the kernel of the map $S \to S_\mathfrak{p}/\mathfrak{p}^n S_\mathfrak{p}$. On the other hand $\mathfrak{q}^{(n)}$ is the kernel of the map $S \to S_\mathfrak{q}/\mathfrak{q}^n S_\mathfrak{q} = S_\mathfrak{q}/\mathfrak{p}^n S_\mathfrak{q}$. Hence it suffices to show that

$$S_\mathfrak{p}/\mathfrak{p}^n S_\mathfrak{p} \longrightarrow S_\mathfrak{q}/\mathfrak{p}^n S_\mathfrak{q}$$

is injective. Observe that the right hand module is the localization of the left hand module by elements $f \in S$, $f \notin \mathfrak{q}$. Thus it suffices to show these elements are nonzerodivisors on $S_\mathfrak{p}/\mathfrak{p}^n S_\mathfrak{p}$. By flatness, the module $S_\mathfrak{p}/\mathfrak{p}^n S_\mathfrak{p}$ has a finite filtration whose subquotients are

$$\mathfrak{p}^i S_\mathfrak{p}/\mathfrak{p}^{i+1} S_\mathfrak{p} \cong \mathfrak{p}^i R_\mathfrak{p}/\mathfrak{p}^{i+1} R_\mathfrak{p} \otimes_{R_\mathfrak{p}} S_\mathfrak{p} \cong V \otimes_{\kappa(\mathfrak{p})} (S/\mathfrak{q})_\mathfrak{p}$$

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where V is a $\kappa(\mathfrak{p})$ vector space. Thus f acts invertibly as desired. $\square$